

Simulation-Based Comparative Benchmarking of QKD Protocols: BB84, E91, and MDI-QKD Over Realistic Optical Fibre Channels

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Abstract—Quantum Key Distribution (QKD) involves the creation of shared secret keys using physics-based principles rather than relying solely on cryptographic methods based on mathematics. The present project focuses on comparing QKD using three different QKD protocols: BB84, E91, and MDI-QKD. The purpose of our study was to evaluate how the performance of these protocols differed in more practical circumstances as opposed to theoretical conditions. We achieved this purpose using the netsquid framework where photon propagation was simulated using a fiber-optic cable subject to photon loss, depolarization noise, and inefficiency of detectors.

Each protocol was evaluated for 30 iterations at a distance of 10 km while varying distances between 1 km to 50 km in order to compare their performances according to increasing distance. Out of all the evaluated protocols, BB84 performed the best by attaining the highest key rates (164 835 bit/sec), while MDI-QKD had the highest overhead due to the 12 times higher number of background communication messages relative to BB84 (51 vs 608 messages). BB84 was also the least vulnerable to attacks on the detector.

MDI-QKD protocol has lower efficiency, but it is also the most secure among all of the analyzed protocols against detector attacks. On the other hand, E91 had no security vulnerabilities because it uses a Bell test for verification of security, but it had the highest losses (72.1%). This occurred due to the fact that entangled photon pairs generated in E91 need to be detected simultaneously by both receivers. Therefore, BB84 protocol is best suited in terms of high speed and efficiency while MDI-QKD is preferable if safety is prioritized, and E91 should be considered if the Bell test is required.

Index Terms—Quantum Key Distribution, BB84, E91, MDI-QKD, QBER, quantum network simulation, Bell inequality, comparative benchmarking, optical fibre channel

I. INTRODUCTION AND LITERATURE REVIEW

The security of traditional cryptographic algorithms such as RSA relies on the computational infeasibility of solving specific mathematical problems. Quantum computers threaten to undermine these assumptions; for example, Shor showed in 1994 that a quantum computer can efficiently perform integer factorisation [1]. Quantum Key Distribution provides an alternative method in which the security is based on

physical principles: an eavesdropper measuring a quantum channel must disturb the state being transmitted and such disturbances will be detectable by Alice and Bob. As the eavesdropper is then identified, the integrity of the quantum communication can be assured [2].

A. BB84

Quantum cryptography was first described in 1984 by Bennett and Brassard [3]. Here Alice encodes each bit in the polarisation of individual photons using either rectilinear $\{|0\rangle, |1\rangle\}$ or diagonal $\{|+\rangle, |-\rangle\}$ basis, choosing randomly. On reception, Bob randomly measures each arriving photon in either of the two bases. Public announcement of the choice of bases and discarding the bits where Alice and Bob used a different basis leaves us with the sifted key. Eve attempting to eavesdrop would inevitably increase Quantum Bit Error Rate (QBER). Mayers [4] and Lo and Chau [5] published a proof against all possible attacks for BB84. This QKD protocol has been experimentally verified across more than 400 km of fibre optics [6].

B. E91

In 1991, Ekert introduced a QKD protocol based on entanglement [7]. An entanglement source creates pairs of photons entangled in Bell states; one photon from the pair is sent to Alice and another to Bob. Alice and Bob measure their photons in any of the three randomly chosen bases. Measurements corresponding to non-identical bases allow testing of the CHSH Bell inequality with the value $S > 2$ indicating entanglement and ruling out any local-hidden-variable attacks [8]. Those measurements done in identical bases become the key. Unlike BB84, E91's security is device-independent; in other words, no assumptions about the nature of the entanglement source were made.

C. MDI-QKD

A tangible weakness of real-world QKD implementations is the detector side-channel attack; due to non-ideal physical

detectors, information may leak out about the measurement results of Bob [9]. This issue has been mitigated by Lo, Curty, and Qi using the introduction of MDI-QKD in 2012 [10]. In this protocol, both Alice and Bob generate quantum states and transmit them to an untrusted third party (Charlie), which then performs BSM and announces the result publicly. Due to the fact that Alice and Bob do not trust Charlie's detectors, there is no way a detector side-channel attack can succeed in MDI-QKD. Experimentally demonstrated at metropolitan level distances, MDI-QKD has become a promising alternative for deploying secure quantum key distribution networks [11].

D. Simulation-Based Protocol Evaluation

Experimental realisations of QKD protocols depend on single-photon sources, precise synchronization, and costly cryogenic detectors, therefore a comparative assessment is unaffordable for many research teams. Instead, simulation offers a feasible option for evaluating the effectiveness of protocols in a repeatable manner. In their study, Doolittle and Cubeddu [12] offer a guide for quantum network simulations and emulations, stressing the importance of physical-layer imperfection modelling. One such example is NetSquid 1.1.x [13], offering a discrete-event quantum network simulator including hardware-precise noise models. While such tools exist, there has been no attempt to benchmark BB84, E91, and MDI-QKD under identical conditions using a single metric. This paper aims to address that gap.

II. METHODOLOGY

A. Simulation Environment

Python 3.12 has been used as a programming language. The Basic Protocol Set comprises all three protocols as independent modules with no quantum-simulator dependencies, thereby allowing rapid multi-run statistics extraction. The Advanced Protocol Set implements the same protocols using the NetSquid 1.1.x discrete-event simulator including hardware-accurate models. All benchmarking results in this paper have been generated through the Basic Protocol Set under identical channel conditions discussed below.

B. Channel and Device Models

Every simulation utilizes the following physics for the quantum communication channel:

- **Fibre loss:** initial fibre loss of 0.1 dB and length-dependent attenuation of 0.2 dB/km (standard SMF-28 fibre) leading to transmission probability $\eta = 10^{-(0.1+0.2 \cdot d)/10}$ for distance d km.
- **Depolarization noise:** bit-flip occurs randomly with probability $p_{\text{depol}} = 0.008$ per qubit passage.
- **Propagation delay:** propagation speed of light in optical fibre $c = 1.9 \times 10^5$ km/s.
- **Detector efficiencies:** $\eta_d = 0.85$ for all single-photon detectors.
- **Entanglement distribution source in E91 protocol:** 0.75 generation efficiency, 0.85 success rate, and 0.92 fidelity in entangled pair generation.

- **MDI-QKD BSM station:** two-photon Bell-State Measurement, success rate $p_{\text{BSM}}(d) = 0.25(1 - 0.005d)(1 - 0.003d)$ with minimum of 0.05.

C. Protocol Implementations

In the **BB84** protocol, the standard four-state prepare-and-measure scheme is utilized. Alice creates 1000 qubits per simulation iteration using randomly selected bit-basis combinations and sending them via quantum channel. Bob performs basis measurement on incoming stream of qubits in random basis, afterwards sifted bits are subjected to QBER estimation, followed by error correction and privacy amplification.

E91 entanglement-based protocol involves creating 1000 entangled state pairs per simulation iteration from the common source situated equidistantly between Alice and Bob. Alice performs measurements in random angle from set $\{0^\circ, 22.5^\circ, 45^\circ\}$, and Bob in random angle from $\{-22.5^\circ, 0^\circ, 22.5^\circ\}$. Measurements are correlated according to $E(\theta_a, \theta_b) = \cos(2(\theta_a - \theta_b))$, which for maximally entangled particles takes into account fidelity of entangled states. The CHSH parameter value equals $S = |E_{00} + E_{02} + E_{22} - E_{20}|$.

MDI-QKD involves creation of 1000 qubits per simulation iteration by both Alice and Bob and their subsequent sending towards the central BSM station Charlie. BSM on both incoming particles is attempted by Charlie, successful measurements (i.e. when two bases are equal) are declared by Charlie and used for sifted key.

D. Benchmarking Protocol

In order to maintain statistical validity, the protocol is run $N = 30$ times independently for the reference distance of 10 km. The mean and standard deviation is calculated per metric. Additionally, a sweep over the distances $\{1, 5, 10, 20, 30, 50\}$ km with $N = 8$ repetitions per distance point is performed in order to explore scalability and metric degradation trends.

E. Performance Metrics

Following each experiment run, the following eight metrics are collected:

- 1) **Key Rate:** length of the sifted key measured in bits after basis reconciliation.
- 2) **QBER:** ratio of incorrect key bits (multiplied by 100%).
- 3) **Latency:** time elapsed in milliseconds during simulation.
- 4) **Channel Loss Rate:** ratio of qubits not received out of total transmitted ($\times 100\%$).
- 5) **Throughput:** rate of secure key generation in bits/second.
- 6) **Communication Overhead:** number of classical messages sent.
- 7) **Synchronization Time:** time to align bases measured in milliseconds.
- 8) **Computation Time per Round:** computation time per qubit in nanoseconds.

III. RESULTS

A. Metric Distributions at 10 km

Table I provides summary statistics of all 8 metrics based on 30 simulation runs at 10 km. Fig. 1 displays the entire distribution of metrics for all 8 protocols in form of boxplots.

B. Key Rate and Throughput

BB84 protocol produces a key rate of 170.2 bits on average per run— $5.7\times$ greater than E91 (29.6 bits) and $8.6\times$ greater than MDI-QKD (19.7 bits). That means that the throughput of BB84 is 164 835 bit/sec compared to MDI-QKD with only 13 850 bit/sec, making $12\times$ throughput difference. The reason for such low key rate of the E91 protocol is due to high losses of the quantum channel: since two photons from the entangled pair should be detected within the same timeframe, the channel transmission probability equals η^2 instead of η . For 10 km distance with $\eta \approx 0.53$, the transmission probability is approximately 0.28, and combined with detector efficiency of 85% leads to channel loss rate of 72.1%.

MDI-QKD's relatively low key rate comes down to the efficiency of the BSM process, which depends on the distance and averages 10%–11% at 10 km in this simulation. Out of a total of 1 000 qubits sent by both parties, only the qubits for which Charlie's BSM is successful and Alice and Bob's bases agree will make up the sifted key.

C. Quantum Bit Error Rate

BB84 and E91 achieve 0.00% QBER at 10 km in this simulation. This is a result of the noise model parameters used: at 10 km, the depolarization rate and the detector error sum up to a value below the error rate that triggers errors in the sifted key with these choices of parameters. MDI-QKD shows 4.33% ($\pm 1.97\%$) mean QBER, due to the interference visibility limitations of Charlie's BSM device (95% in the current model) and the distance-dependent error rate of Charlie's detectors. The MDI-QKD QBER does not reach the security threshold of 11%, beyond which privacy amplification becomes impossible using conventional BB84-like security proofs.

D. Classical Communication Overhead

The biggest functional distinction between these three protocols is the classical communication overhead involved in their implementation. BB84 takes 51 messages per run for session establishment, basis reconciliation, error correction, and key confirmation. E91 requires 131 messages – the additional cost of three-party entanglement synchronization and Bell-test coordination, plus more complicated post-processing because of the entanglement-based security proof. MDI-QKD requires a huge number of 593 messages per run, $12\times$ higher than BB84. This is caused by Charlie's BSM announcement (one message per successful measurement), three-party synchronization, and additional verification rounds.

E. Performance-Security Trade-Off

The key trade-off is illustrated in Fig. 2. Each point represents a single run, and diamond markers represent protocol-wise means. BB84 performs with high throughput and very low QBER, whereas E91 is located closer to mid-throughput and zero QBER, and finally, MDI-QKD achieves lowest throughput, but with the best practical security against any possible detector side channel attack, irrespective of QBER level.

F. E91 Bell Parameter

The average CHSH value calculated over all runs was $S \approx 2.60$, above the threshold value of $S = 2$ for nonlocal realism and below the theoretical maximum for quantum correlations ($S = 2\sqrt{2} \approx 2.83$). This ensures that the entangled states prepared within the simulation protocol have sufficient fidelity, providing the required security guarantee based on quantum mechanics. The remaining difference is determined by both the 92% entanglement fidelity of the source model and depolarisation noise during transmission.

G. Distance Dependence

The evolution of eight metrics at 1–50 km is shown in Fig. 3. The protocols demonstrate varying performance dependence on distance. Raw key rate declines linearly for all three protocols. BB84 is the least vulnerable to distance degradation, maintaining positive key generation even beyond 50 km. As for E91, coincidence detection leads to higher sensitivity, causing key rate to reach 0 faster than for other protocols; after 30 km, many runs produce insufficient coincidences to generate keys. MDI-QKD lies somewhere in-between, as it has the effect of halving the distance for each photon owing to the mid-point relay, but suffers due to BSM efficiency penalty at large distances.

H. Normalised Comparison

Fig. 4 shows a radar chart where all measures have been normalised to $[0, 1]$ and plotted such that further out = better. Measures for which lower values are desired (QBER, loss rate, overhead, latency) are inverted first. BB84 prevails on all axes of efficiency; E91 wins on latency and QBER; MDI-QKD is lagging behind in most aspects but is unique in guaranteeing full detector-independence against side channels without making any assumptions about the measurement devices.

IV. DISCUSSION

Based on the results, we can formulate the following three criteria for choosing between the protocols:

(1) High-speed, trusted devices. BB84 is the right choice. Simplicity leads to the highest rate, lowest overhead, and fastest computational performance per round. If detectors are protected in a network where BB84 runs, its high efficiency will outweigh its inability to deliver a detector-independent guarantee.

TABLE I
PERFORMANCE METRIC SUMMARY: MEAN $\pm$ STANDARD DEVIATION (30 RUNS, 10 KM FIBRE)

Metric	BB84	E91	MDI-QKD
Raw Key Rate (bits)	170.2 $\pm$ 11.2	29.6 $\pm$ 3.7	19.7 $\pm$ 2.3
QBER (%)	0.00 $\pm$ 0.00	0.00 $\pm$ 0.00	4.33 $\pm$ 1.97
Latency (ms)	1.03 $\pm$ 0.02	1.05 $\pm$ 0.05	1.43 $\pm$ 0.13
Channel Loss Rate (%)	38.5 $\pm$ 1.3	72.1 $\pm$ 1.5	39.7 $\pm$ 1.3
Throughput (bits/s)	164 835 $\pm$ 10 324	28 373 $\pm$ 3 495	13 850 $\pm$ 1 751
Communication Overhead (msgs)	51 $\pm$ 0	131 $\pm$ 0	593 $\pm$ 12
Synchronisation Time (ms)	1.26 $\pm$ 0.00	1.52 $\pm$ 0.00	2.85 $\pm$ 0.00
Computation Time/Round (ns)	14.8 $\pm$ 5.1	387.9 $\pm$ 25.4	67.3 $\pm$ 21.7

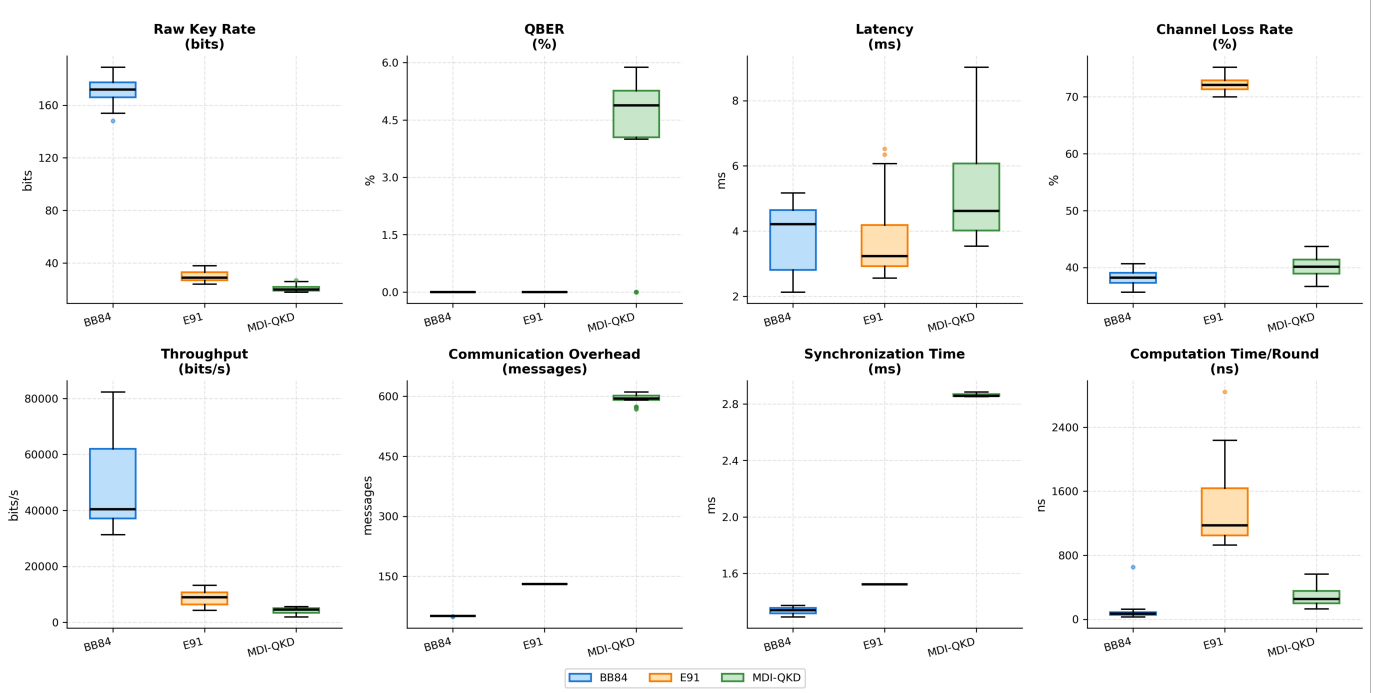


Fig. 1. Box-plot distributions of all eight performance metrics across 30 runs per protocol at 10 km. Horizontal lines inside boxes denote medians; boxes span the IQR; whiskers extend to $1.5 \times \text{IQR}$.

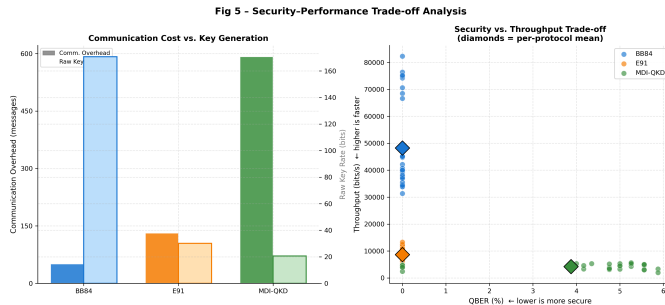


Fig. 2. Security-performance scatter diagram: QBER (lower = more secure, x-axis) vs. throughput (higher = faster, y-axis). Each point represents a single run, and diamond markers are protocol-wise means.

(2) **Need for quantum mechanical proof of security.** E91 is selected if the protocol needs to provide a quantum-

mechanical security proof based on a verifiable physical principle (violation of Bell inequalities) and not on the security of Alice's and Bob's devices. This approach requires paying a price in terms of a $5.7 \times$ drop in key generation rate compared to BB84 and relatively high loss rate in the channel.

(3) **Untrusted or adversarially controlled measurement devices.** In order to counter potential detector side-channel attacks, only MDI-QKD is the solution in the case where an adversary controls or has access to measurement devices. The $12 \times$ overhead and $8.6 \times$ reduction in key generation rate compared to BB84 can be considered as acceptable trade-off price.

One caveat regarding this study is that while it accounts for losses and depolarising noise in the photon path, it does not account for phase noise, timing jitter, and multi-photon attacks. These factors should be considered in future research and compared together with twin-field QKD technique [14].

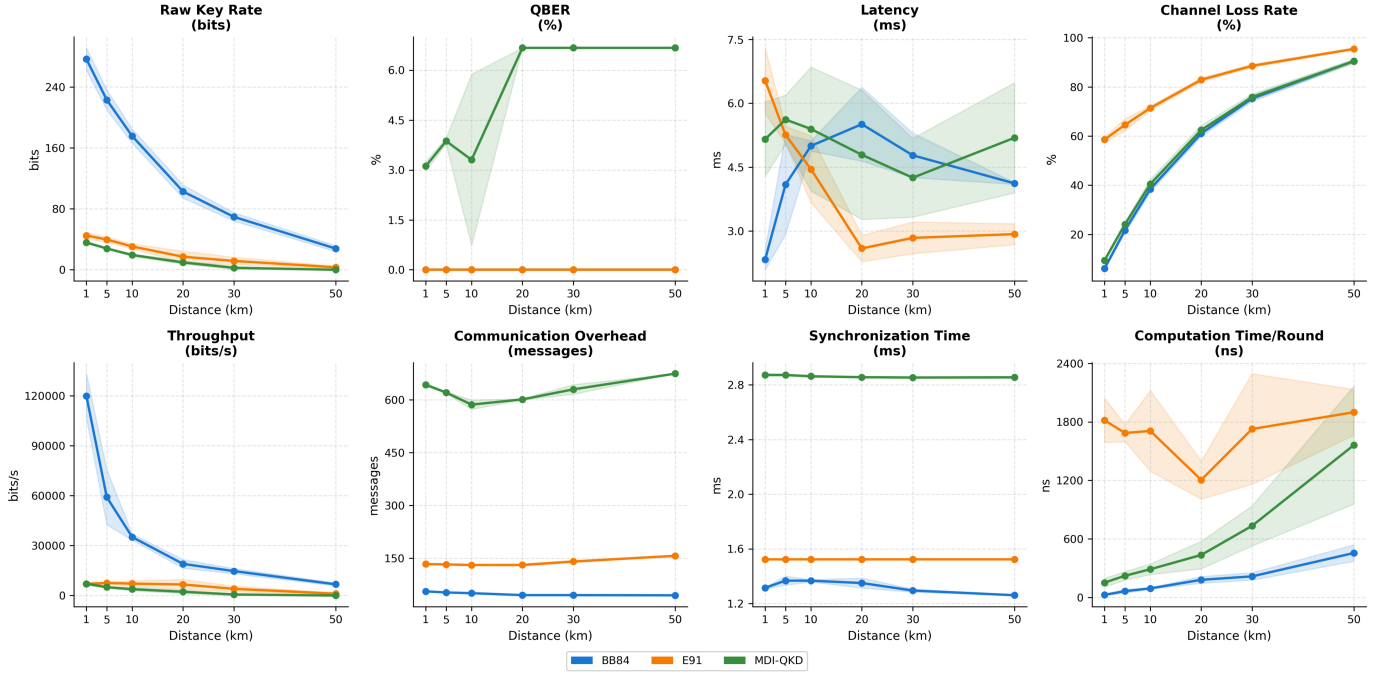


Fig. 3. Distance analysis: average value of the metric over 8 simulations at each distance point. Standard deviation is marked by shaded areas. Channel models are identical across all protocols.

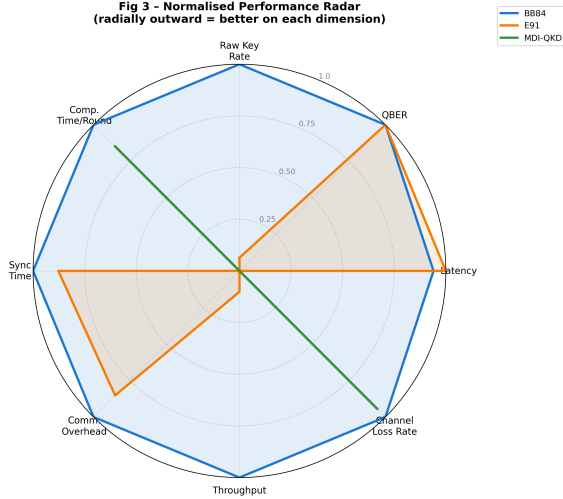


Fig. 4. Performance comparison normalised radar chart. Values on axes are rescaled within [0, 1], where higher means better performance. Inverted axes are: QBER, channel loss rate, communication overhead, latency, synchronisation time, and computation time.

V. CONCLUSION

In this paper, a simulation benchmark comparison of the three QKD protocols was conducted using identical realistic optical fibre channel parameters. Thirty independent simulation iterations at a 10 km distance level and a six-point distance sweep between 1 km and 50 km demonstrate enough statistical significance for the following conclusions:

- The BB84 protocol exhibits the most efficient re-

sults, which include: maximum 170.2 raw key rate, 164 835 bits/s throughput, and minimum 51 classical communication messages per session.

- The high value of the Bell inequality $S \approx 2.60$ in E91 is a strong indication that entanglement-based security is achieved, although coincident-detection requires 72.1% of channel losses at 10 km.
- MDI-QKD avoids detector-side channel attacks but requires 593 classical messages per session and $12\times$ less throughput than BB84.
- Protocols degrade with the distance increase predictably, while MDI-QKD demonstrates the steepest curve starting from 20 km because of accumulated photon losses and decreased probability of BSMs.

These results may serve as a basis for the rational selection of one of the three dominating protocols during quantum networks' design.

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